Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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Abstract

Ocean source separators normally infer a changing background only from the mixture being separated. We ask whether a strictly causal estimate of the expected background, computed before the mixture arrives, provides useful additional evidence. FC-BSR predicts a current spectrum from historical one-third-octave levels and injects it into BS-RoFormer band tokens through concatenation, feature-wise linear modulation (FiLM), or cross-attention. We separate three claims through a staged evaluation. First, a controlled surrogate-background study with exact stems selects FiLM: on 768 paired six-family examples it improves SI-SDR improvement (SI-SDRi) over an unconditioned separator by 0.34 dB, while JS divergence alone does not predict gain. Second, we replace the surrogate with 42 hours of continuous raw hydrophone background, retain synthetic targets so exact stems remain available, and compare 5-s, 30-s, 1-min, and 10-min windows. All four temporal priors improve separation on 1,024 held-out clips; 5 s gains +1.226 dB ($[1.082, 1.381]$). However, learned forecasting beats persistence only at 5 s, so this stage supports time-correlated conditioning more strongly than a generally superior forecaster. Third, on 1,225 annotated real whale calls with no isolated stems, a matched reference-free comparison lowers residual event probability from .523 to .507 while leaving event-stem evidence unchanged; the off-event change rises from .0956 to .1022. The forecast prior therefore improves real-event semantic removal, but not every fidelity criterion.

1 Introduction

Passive acoustic backgrounds vary with shipping, weather, biology, and season. A separator that only observes the current mixture must estimate both the target and that background at once. Long-context soundscape forecasting offers an additional, strictly causal signal: what the background was expected to look like immediately before the mixture arrived. Our hypothesis is that this prior helps a separator reject predictable energy while retaining an unexpected injected source.

We combine the band-split RoPE Transformer (BS-RoFormer) architecture [4] with a causal recurrent forecast encoder. The evidence is deliberately ordered from control to realism. The first study uses a TOL-derived background surrogate and exact source stems to select the injection mechanism and probe source diversity. The second replaces only the background with continuous raw hydrophone audio, retaining exact references through synthetic target injection. The third replaces the test target with annotated real whale calls; because isolated stems no longer exist, it uses semantic and partial-ground-truth checks instead of SI-SDR. Each stage removes one simplifying assumption, and its conclusion is restricted to the question it can identify.

The contributions are therefore: a differentiable forecast-conditioning interface with matched concatenation, FiLM, and cross-attention ablations; a leakage-audited three-stage protocol reporting forecast accuracy, paired separation, semantic removal, preservation, uncertainty, and runtime; a six-family generator with spectral-shift and entropy analysis; and a usable local application for inspection, playback, and download.

2 Data and evaluation ladder

The three studies use different observations by design. "Real background" means that the noise waveform is recorded rather than reconstructed; "real target" means that the event itself is recorded. These properties must not be conflated. The first study has synthetic background and targets with complete ground truth, the second has real background and synthetic targets with complete ground truth, and the third has real background and real targets but no separated ground truth.

We use NOAA–Navy SanctSound hourly one-third-octave level (TOL) products [5]. MB03 deployments 01 and 02 form the training set, 03 is validation, and 04 is held-out test. A sample at hour t is eligible only when all H preceding hours and hour t are finite and exactly contiguous. Normalization statistics are fitted on the two training deployments only.

The public TOL product is an aggregate and cannot determine waveform phase. For each current spectrum S_t, we interpolate its dB envelope on FFT bins, draw a seeded random phase and Rayleigh-like magnitude jitter, invert the FFT, and normalize RMS. This gives background surrogate n_t. It preserves the coarse measured spectral envelope but not transients, phase, or within-hour events. It must not be described as recovered raw audio.

For the field experiment we use the corresponding continuous 8-kHz SanctSound hydrophone recordings directly: 6 hours each from MB03_01, 02, and 03, and a continuous 24-hour MB03_04 test day (42 hours total). One-second Hann periodograms are integrated into 23 one-third-octave

bands from 25 to 4000 Hz, averaged to 5-s features, and power-aggregated to 30 s, 1 min, or 10 min. No waveform phase is synthesized in this experiment.

The original injected signal s_t follows the controllable synthesis design developed in MINUS: harmonic partials plus band-limited noise, attack/release edges, and low-frequency amplitude modulation. The broad profile adds chirps, inharmonic tones, impulse trains, frequency-modulated tones, and band-limited noise, while expanding carrier, modulation, envelope, transient, and noise-mix ranges. Parameters, family, and source-to-background SNR are drawn deterministically for every sample. The mixture is $x_t = n_t + s_t$, and clipping is prevented by common rescaling.

The third stage uses a separate, strongly annotated 2-kHz DCLDE 2013 right-whale recording set. Its day-disjoint construction and reference-free metrics are defined in Section 6; it is not pooled with SanctSound.

3 Method

Let $S_{t-H:t-1} \in \mathbb{R}^{H\times F}$ be normalized historical TOL. A GRU produces

$$z_t = \mathrm{LN}(\mathrm{GRU}(S_{t-H:t-1})), \tag{1}$$

$$\widehat{S}_t = g_\theta(z_t). \tag{2}$$

Only observations ending at $t-1$ enter the encoder. All three injection ablations use $c_t = \widehat{S}_t$; the baseline omits the forecaster.

BS-RoFormer converts mixture STFT bins into band tokens $X \in \mathbb{R}^{B\times T\times K\times D}$. Let $P_k(c_t)$ be a learned band-specific projection and $E(c_t)$ a sequence of embedded forecast bins. We compare three residual injection operators at the same point before the time/frequency Transformer:

$$\text{Concat: } \widetilde{X}_k = X_k + f([X_k; P_k(c_t)]), \tag{3}$$

$$\text{FiLM: } \widetilde{X}_k = (1+\tanh\gamma_k(c_t)) \odot X_k + \beta_k(c_t), \tag{4}$$

$$\text{Cross-attn: } \widetilde{X} = X + \mathrm{Attn}(X, E(c_t), E(c_t)). \tag{5}$$

The residual branches are initialized to zero, so every run begins from the same audio mapping. Concatenation doubles the local feature width before a fusion MLP; FiLM directly changes band-token gain and offset; cross-attention learns content-dependent alignment to forecast-frequency tokens.

For the controlled surrogate-background study, the end-to-end objective is

$$\mathcal{L} = \mathcal{L}_{\text{wave}} + \tfrac{1}{2}\sum_r \mathcal{L}_{\text{complex STFT},r} + \lambda\|\widehat{S}_t - S_t\|_1, \tag{6}$$

with $\lambda = 0.2$ for conditioned variants. In that first study, gradients from both objectives reach the forecast encoder. In the continuous-waveform and real-event studies, the predictor is instead selected by validation forecast error and then frozen before separator training. This deliberate difference prevents the separation loss from degrading an independently measured forecast, while preserving the same conditioning interface.

4 Controlled surrogate study

This first evidence tier asks which conditioning operator works under complete ground truth. It is a mechanism-selection experiment, not yet evidence about raw ocean waveforms or real target events.

All variants use identical samples, synthesis seeds, optimizer schedule, model width, matched separator-core initialization, and checkpoint rule. The checkpoint with maximum validation SI-SDR improvement (SI-SDRi) is evaluated once on MB03_04. SI-SDR follows the scale-invariant projection form; improvement subtracts mixture SI-SDR. We also report residual-background SI-SDR, forecast MAE in dB, inference real-time factor (RTF), parameter count, and the fraction of paired test examples where conditioning beats the baseline. Metric uncertainty is a seeded paired bootstrap over test examples. BSS evaluation background is discussed by Vincent et al. [6].

For distribution shift, waveform power is pooled into $K = 64$ log-frequency bins and normalized to p. Let q be the mean distribution of 512 draws from the original narrow harmonic generator. We report the directional $D_{\mathrm{KL}}(p\|q)$ and use the finite symmetric quantity

$$D_{\mathrm{JS}}(p,q) = \tfrac{1}{2}D_{\mathrm{KL}}(p\|m) + \tfrac{1}{2}D_{\mathrm{KL}}(q\|m), \quad m = (p+q)/2, \tag{7}$$

as the primary shift metric [3]. Because distance from the old generator need not measure source complexity, we also report normalized spectral entropy $H(p)/\log_2 K$ and source–ocean histogram intersection $\sum_k \min(p_k, n_k)$.

Table 1: Matched 2,000-update broad-source ablation.

Injection	Val. SI-SDRi (dB)	Test SI-SDRi (dB)	Paired Δ_0 (dB)	Params (k)
None	4.96	5.00	0	231
Concatenation	5.17	5.30	+0.15	238
FiLM	**5.35**	**5.39**	**+0.34**	255
Cross-attention	4.99	5.11	+0.01	220

4.1 Conditioning ablation and selection

Each model uses 4,000 examples per training deployment, 2,000 updates, and SNR drawn uniformly from $[-12, 12]$ dB. We select only by MB03_03 validation SI-SDRi: FiLM obtains 5.35 dB, versus 5.17 for concatenation and 4.99 for cross-attention. On the ordinary 256-example broad MB03_04 test, their scores are 5.39, 5.30, and 5.11 dB, respectively; the unconditioned model reaches 5.00 dB. Thus the held-out test confirms, rather than determines, the choice.

At fixed 0 dB, FiLM gains 0.34 dB over the same unconditioned outputs ($[0.21, 0.48]$), concatenation gains 0.15 dB ($[0.02, 0.27]$), and cross-attention gains 0.01 dB ($[-0.08, 0.10]$). FiLM is a good fit because a background

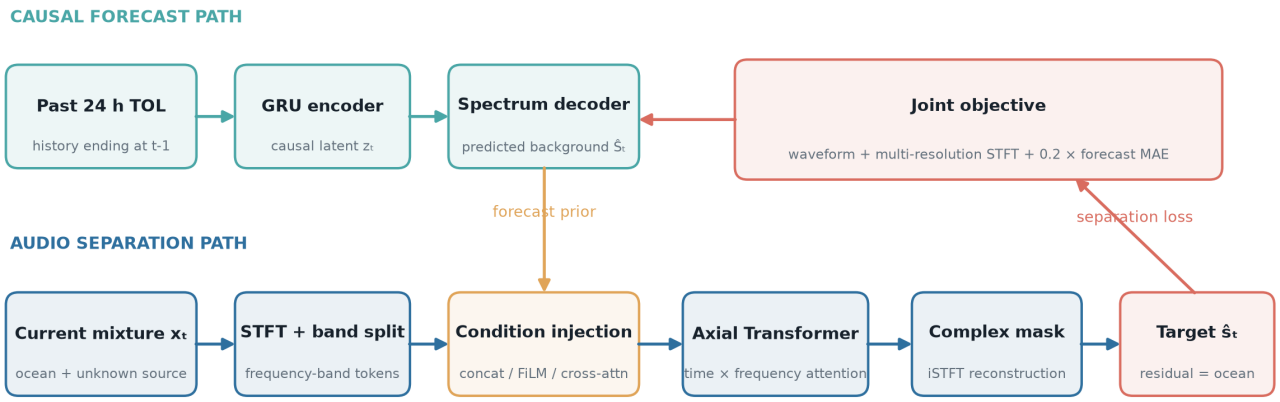


Figure 1: FC-BSR architecture. Historical TOL ending at $t-1$ is encoded into a forecast of the current ocean spectrum. Concatenation, FiLM, and cross-attention inject that prior before the axial Transformer. The controlled study trains both paths jointly; the two real-waveform studies freeze a validation-selected predictor before training the separator. The unconditioned baseline omits the forecast path.

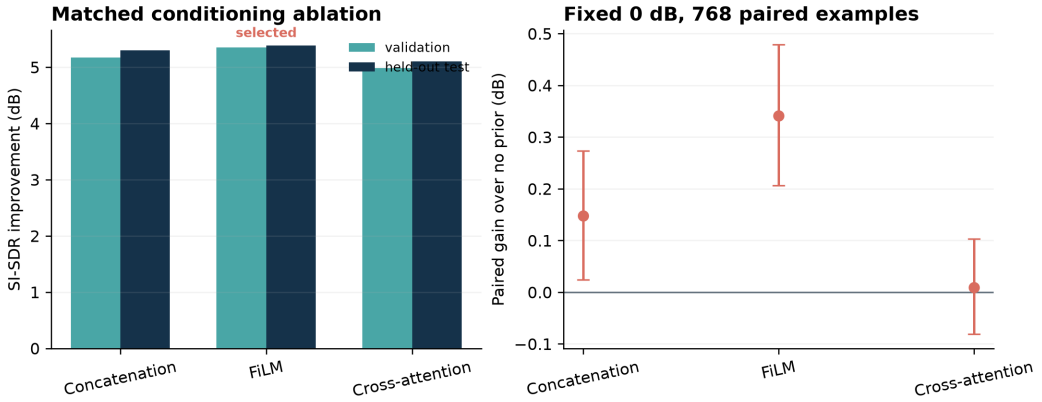


Figure 2: Matched injection ablation. Model selection uses MB03_03 validation only (left). On 768 fixed-0-dB paired MB03_04 examples (right), FiLM has the largest gain and its bootstrap interval excludes zero; cross-attention does not.

spectrum naturally specifies which band features to attenuate, amplify, or shift. It preserves the audio evidence while applying a low-depth multiplicative/additive prior. Concatenation must relearn a compression of doubled features, while cross-attention must estimate an alignment from only a coarse hourly spectrum; its extra flexibility is not rewarded at this data scale. This explanation is mechanistic and consistent with the results, but is not a causal proof.

Input SNR (dB)	Cond. SI-SDR (dB)	Cond. SI-SDRi (dB)	Base SI-SDRi (dB)
-12	-4.63	7.30	7.32
-6	1.00	6.95	6.78
0	5.85	5.81	5.40
+6	9.55	3.51	2.94
+12	11.94	-0.10	-0.70

Table 2: Separation by injected-source strength (128 examples per level).

At weak and equal-strength inputs the selected FiLM model improves the mixture by at least 5.81 dB. At +12 dB, SI-SDRi is near zero because the mixture is already target-dominant, while absolute target SI-SDR remains 11.94 dB. This motivates separate improvement and absolute-quality gates rather than one threshold for all input strengths.

4.2 Broad-source distribution study

To isolate source shape from strength, the main diversity evaluation fixes SNR at 0 dB and uses 128 held-out MB03_04 examples from each family. The resulting 768 examples are exactly paired in ocean waveform, source waveform, timestamp, and SNR.

Across all broad examples the baseline reaches 5.36 dB and FiLM reaches 5.70 dB, a paired gain of 0.34 dB. The gain–JS correlation is -0.085; the gain falls from 0.54 dB in the lowest JS quintile to 0.00 dB in the highest. Thus "farther from the original generator" is not a sufficient account of when an ocean prior helps.

Spectral entropy is more diagnostic. The gain–entropy correlation is 0.145; the highest entropy quintile gains +0.80 dB ($[0.47, 1.17]$). By family, band-limited noise gains +0.54 dB ($[0.19, 0.92]$), impulse trains gain 0.47 dB ($[0.16, 0.79]$), and frequency-modulated targets gain 0.43 dB ($[0.08, 0.82]$). FiLM therefore uses the prior broadly, with its strongest benefit on diffuse sources; raw JS/KL distance is not the governing variable.

5 Sub-hour real-background study

This second evidence tier replaces the reconstructed background with continuous raw audio while retaining injected synthetic targets. It can therefore test separation with exact signal and noise references in a real soundscape, but it does not yet test generalization to naturally occurring targets.

The experiment fixes history coverage at one hour, giving 720, 120, 60, or 6 causal history steps. For target window t, the model receives only $S_{t-H:t-1}$; the background target is the actual waveform N_t. A broad synthetic source is injected into a clip inside that window, $x_t = N_t + \alpha s_t$, so clean signal and noise references remain exact. Absolute clip times and injection seeds are shared across resolutions.

The GRU is first trained for up to 2,500 updates. Because short-window ocean spectra are highly persistent, its zero-residual initialization is the explicit step-0 persistence candidate; a learned checkpoint is retained only if it lowers MB03_03 forecast MAE. The selected predictor is then frozen while the FiLM separator trains for at least 3,000 and at most 5,000 updates, with the best MB03_03 SI-SDRi checkpoint evaluated once on MB03_04. Thus prediction quality cannot be silently degraded to optimize separation.

All four conditioned models pass the five-SNR engineering gate and run at about RTF 0.0024. The 5-s window has the highest absolute SI-SDRi and paired gain, but the four gain intervals overlap, so the experiment does not establish a significant ranking among resolutions. The important negative result is that more expressive prediction is not automatically useful: only 5 s improves over persistence on validation (by 1.4%), while 10-min training has only 60 distinct next-window examples and strongly overfits. Temporal background information is useful; with the present data, persistence supplies most of that information.

6 Reference-free evaluation on real events

The third evidence tier replaces the synthetic test target with a naturally occurring annotated event. Real event recordings do not provide isolated source and background stems, so SI-SDR is unavailable and waveform fidelity cannot be fully identified. This stage instead asks the narrower question of whether forecast conditioning improves semantic event removal without an obvious preservation failure. We use the DCLDE 2013 North Atlantic right-whale corpus [1]: continuous 2-kHz mono WAV files are divided into 8-s clips, and recording days—not clips—define the split. Certain upcall annotations yield positives; negatives exclude every certain or possible call plus a 2-s margin. The balanced manifest contains 5,292 training, 2,696 validation, and 2,676 test clips. Only clean training/validation backgrounds receive synthetic, time-localized upcalls for supervised separator training. For the matched conditioning experiment, each target receives eight preceding 8-s, 30-band spectra; file-start clips without a full 64-s causal history are excluded from both arms. This leaves 1,225 real positive test clips, none of which participate in checkpoint selection. The residual GRU forecast lowers validation MAE from the persistence value of 2.081 to 1.921 dB (7.67%). It is then frozen. BS-RoFormer and FC-BSR share samples, seeds, transferred weights, 2,500-update budgets, and validation-only checkpoint selection; FC-BSR alone receives the forecast through FiLM. Their controlled validation SI-SDRi values are 13.871 and 13.925 dB, respectively.

Following learned referenceless separation assessment [2] and recent semantic reference-free metrics [7], we train a separate event classifier, calibrate its temperature on validation, and freeze it. Test AUROC is 0.9704, average precision 0.9738, and F1 0.9196. For event estimate $\hat{e}$ we enforce exact mixture consistency, $\hat{n} = x - \hat{e}$, then intervene with $y_g = \hat{n} + 10^{g/20}\hat{e}$ for $g \in \{-60, -30, -24, -18, -12, -6, 0\}$ dB. A credible output should reduce event probability in $\hat{n}$, retain event evidence in $\hat{e}$, recover that evidence monotonically as g rises, and minimally change samples outside annotated intervals. Identity ($\hat{e} = 0$) and all-to-event ($\hat{e} = x$) controls guard against gaming any one score.

FC-BSR lowers background event probability by a further 0.0160 relative to BS-RoFormer; a paired bootstrap over the three recording-day means gives $[-0.0210, -0.0132]$. Event-stem probability is unchanged within uncertainty (-0.00035, $[-0.00283, 0.00552]$), as is monotonicity (-0.00109, $[-0.00342, 0.00265]$). The stronger suppression has a small but consistent preservation cost: off-event relative change rises by 0.00662 ($[0.00331, 0.00781]$), and energy outside annotated inter-

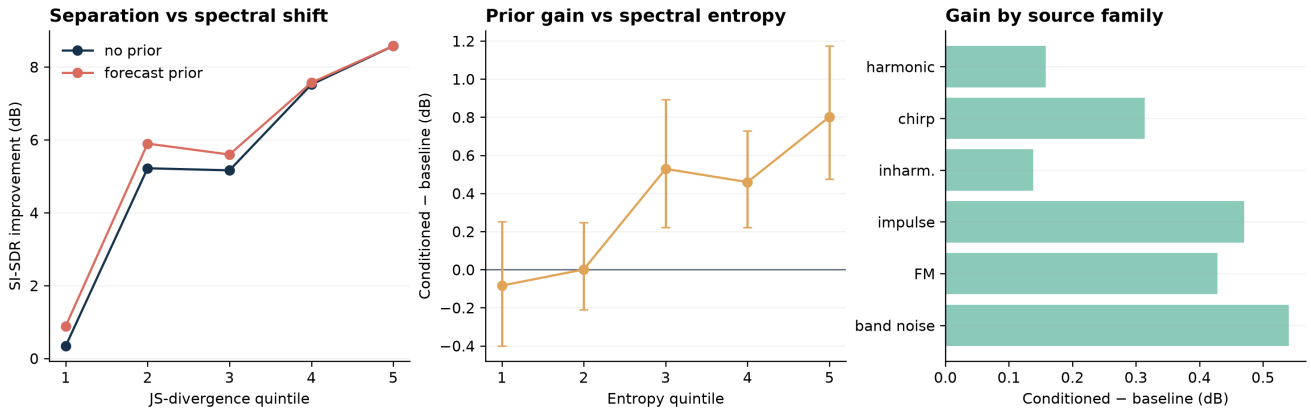


Figure 3: Broad-source results at 0 dB input SNR. Left: both models improve as JS-shift quintile increases, but the forecast prior does not close the gap. Middle: the prior becomes beneficial only in the most spectrally diffuse entropy quintile; bars show paired 95% bootstrap intervals. Right: family-wise FiLM gains are largest for band noise, impulses, and FM targets.

Table 3: Sub-hour prediction and separation on continuous real ocean audio. Intervals are paired 95% bootstrap intervals over 1,024 held-out clips.

Window	History steps	Selected forecast	Val. MAE (dB)	Cond. SI-SDRi (dB)	Base SI-SDRi (dB)	Paired gain (dB)	95% interval (dB)
5 s	720	GRU, step 1600	1.366	**29.753**	28.527	**+1.226**	[1.082, 1.381]
30 s	120	persistence	0.782	29.630	28.534	+1.095	[0.935, 1.265]
1 min	60	persistence	0.687	29.661	28.533	+1.128	[0.971, 1.298]
10 min	6	persistence	1.371	29.711	28.535	+1.176	[1.036, 1.326]

Table 4: Matched model comparison on 1,225 real calls. Arrows indicate the preferred direction; neither model has separated real-event ground truth.

Method	$p(x)$	$p(\hat{n})\downarrow$	$p(\hat{e})\uparrow$	Mono.↑	Off-event↓
BS-RoFormer	.855	.523	.901	.986	.0956
FC-BSR	.855	**.507**	.901	.985	.1022

vals rises by 0.0101 ($[0.00310, 0.01574]$). Identity and all-to-event controls still reject the two trivial solutions: the former removes nothing, while the latter changes all off-event audio. Thus the forecast prior improves real-event semantic removal without a detectable reduction in event-stem evidence, but it does not yield an unqualified improvement in source fidelity. This distinction is only visible because the semantic and partial-GT preservation criteria are reported together; the classifier does not substitute for unavailable separated stems.

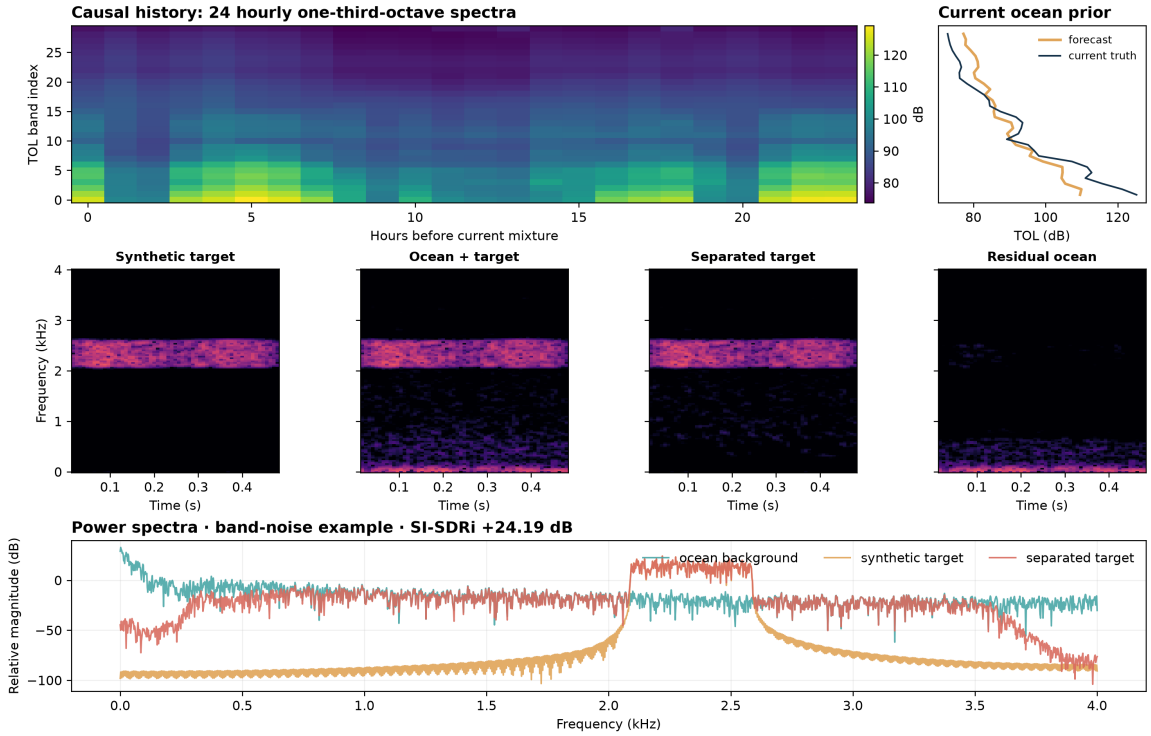


Figure 4: Representative high-entropy band-noise example. The causal TOL history and current forecast are shown above; source, mixture, separated target, and residual-ocean spectrograms are shown in the middle; spectra are overlaid below. The example illustrates how the selected FiLM prior removes low-frequency ocean energy while retaining a diffuse band target.

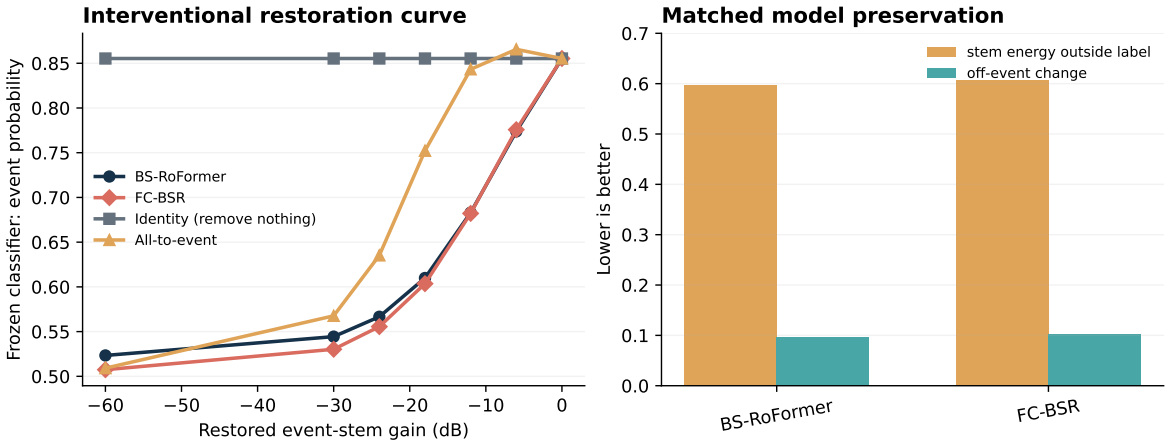


Figure 5: Real-event evaluation without separated ground truth. Left: FC-BSR suppresses the event slightly more strongly than matched BS-RoFormer while both restore it nearly monotonically; identity and all-to-event audit the metric. Right: FC-BSR's stronger semantic removal comes with a small preservation cost.

7 Interpretation and usability

The three evidence tiers support nested, not interchangeable, conclusions. The surrogate study selects FiLM and establishes controlled full-reference gain. The continuous-waveform study shows that a causal temporal prior helps on unseen real backgrounds, but the selection of persistence at three of four resolutions prevents a general claim that learned dynamics are necessary. The real-event study shows semantic transfer beyond matched BS-RoFormer, while its preservation cost and missing stems prevent an unconditional fidelity or deployment claim.

The engineering gate requires overall SI-SDRi of at least 5 dB, at least 3 dB improvement for input SNR at or below 0 dB, at least 8 dB absolute target SI-SDR for stronger inputs, and frontend RTF below one. The conditioned model passes all four checks. Its batched RTF is 0.0040; the complete chunked frontend processes a 10-second WAV at RTF 0.043 on an RTX 3060. These checks establish computational practicality, not field readiness. Field readiness still requires multi-site, multi-season evidence and either isolated real stems or stronger partial references.

8 Local application

The repository includes the validation-selected one-megabyte FiLM checkpoint and a dependency-light local web application. A user supplies the last 24 hourly TOL rows and a WAV mixture. The application validates and normalizes history, predicts the current spectrum, separates overlapping waveform chunks, and exposes target and residual-ocean audio for playback and download. It also plots the predicted spectrum and reports duration, latency, and RTF. Requests stay on localhost; no recording or history is sent to a remote service.

9 Limitations

The real-waveform experiment removes the surrogate-background limitation, but the injected targets remain synthetic and all recordings come from one site. Only 12 training hours are available; at 10-min resolution this leaves 60 distinct next-window examples, which is inadequate for a flexible GRU. Multiple recording sites, seasons, training seeds, and real vessel, biological, and sonar targets are needed before deployment claims can be made.

The six source families are deliberately more diverse than the original generator, but JS divergence and entropy remain spectral summaries: neither captures semantic identity, long temporal organization, or recording-device effects. Family-wise findings should be confirmed with real vessels, animals, sonar, weather, and anthropogenic events before selecting a production model.

10 Conclusion

FC-BSR provides a concrete, leakage-resistant route from soundscape history to source separation and a usable local interface. The staged evidence yields three distinct conclusions. Controlled complete-reference experiments select FiLM and show a +0.34 dB paired gain. Continuous real-background experiments show that causal temporal conditioning improves separation at all four tested resolutions, with a +1.226 dB gain at 5 s; because learned dynamics beat persistence only at 5 s, they do not establish that a complex forecaster is generally necessary. Finally, annotated real calls show stronger semantic removal than matched BS-RoFormer without detectable loss of event-stem evidence, but with slightly greater off-event change. Forecast conditioning is therefore useful across progressively more realistic tests, yet the present evidence does not support an unconditional fidelity or deployment claim.

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